Counterexamples to Erdős Problem 1075

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Abstract

Erdős conjectured that, for every fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$, every sufficiently large r-uniform hypergraph on N vertices with at least $(1+\varepsilon)(N/r)^r$ edges contains a subgraph on $m \to \infty$ vertices with at least $c_r m^r$ edges. We give counterexamples for every $r \geq 5$. The base construction is a sequence of explicit 5-uniform hypergraphs G_n whose unnormalized Lagrangians satisfy

$$5^{-5}\left(1+\frac{1}{4(3n)^3}\right) \leq \lambda(G_n) \leq 5^{-5}+\frac{1}{4!n}.$$

Blow-ups give the counterexample for $r = 5$, and adjoining common vertices to every edge lifts it to all larger uniformities. Thus Problem 1075, as a statement for all $r \geq 3$, has a negative answer. The endpoint question remains open for $3 \leq r \leq 4$, including the case 2/9 for 3-graphs. The proof is formalized and verified in Lean 4.

1 Introduction

All hypergraphs in this paper are finite and simple. For a hypergraph H, write $v(H) = |V(H)|$, $e(H) = |E(H)|$, and let $H[S]$ denote the subgraph induced by $S \subseteq V(H)$.

Erdős proved that, for every fixed $r \geq 2$ and $\varepsilon > 0$, an r-uniform hypergraph on N vertices with at least εN^r edges contains a complete r-partite r-uniform hypergraph with equal part sizes tending to infinity with N [1]. In particular, it contains a subgraph on $m \to \infty$ vertices with $(m/r)^r$ edges. Erdős subsequently conjectured that, for each fixed $r \geq 3$, there is a constant $c_r > r^{-r}$ such that, for every $\varepsilon > 0$ and all sufficiently large N, the assumption

$$e(H) \geq (1+\varepsilon)(N/r)^r$$

forces a subgraph on $m = m(N) \to \infty$ vertices with at least $c_r m^r$ edges [3, p. 80]. This is Problem 1075 in Bloom's collection [13].

Use the density $d(H) = e(H)/\binom{v(H)}{r}$. A number $\alpha \in [0,1)$ is a *jump* for r-graphs if there is a $c > 0$ such that, for every $\varepsilon > 0$ and integer $m \geq r$, all sufficiently large r-graphs of density at least $\alpha + \varepsilon$ contain an m-vertex subgraph of density at least $\alpha + c$; otherwise it is a *non-jump* [4]. The threshold $(N/r)^r$ corresponds asymptotically to $\alpha_r = r!/r^r$. Erdős proved that every number in $[0, \alpha_r)$ is a jump for r-graphs [2]. Thus Problem 1075 is the endpoint question at α_r. Frankl and Rödl disproved the earlier jumping constant conjecture by constructing non-jumps for every $r \geq 3$ [4, Theorem 1.2]. Frankl, Peng, Rödl and Talbot later showed that $(5/2)r!/r^r$ is a non-jump for every $r \geq 3$ [5], and Peng proved the lifting theorem that carries a non-jump of the form $c\,r!/r^r$ to all larger uniformities [6, Theorem 1]. Yan and Peng proved that 12/25 is a non-jump for 3-graphs, yielding $(54/25)r!/r^r$ for every $r \geq 3$ by lifting [7].

For $r \geq 4$, Shaw proved that $2r!/r^r$ is a non-jump and is the smallest non-jump obtainable from his finite-pattern formulation of the Frankl–Rödl method [8, Theorems 4.3 and 4.4(ii)]. For $r = 3$, the corresponding threshold is $6(5\sqrt{5} - 2)/121$ [8, Theorem 4.4(i)]. Liu and Mubayi subsequently proved that $4/9$ is a non-jump for 3-graphs, below this threshold [9]. The endpoint $2/9$ for 3-graphs remains open; Baber and Talbot proved, using flag algebras, jump intervals above it, the first beginning at 0.2299 [10]. The result below shows that the endpoint is a non-jump for every $r \geq 5$; the cases $3 \leq r \leq 4$ remain open.

Theorem 1. *For every $\gamma > 5^{-5}$, there is an $\varepsilon > 0$ and there are arbitrarily large 5-uniform hypergraphs H such that*

$$e(H) \geq (1+\varepsilon)\big(v(H)/5\big)^5, \qquad e(H[S]) < \gamma |S|^5 \quad \textit{for every nonempty } S \subseteq V(H).$$

For every nonempty vertex set S, any subgraph on S has at most $e(H[S])$ edges, so the theorem bounds all subgraphs on that vertex set. Taking $5^{-5} < \gamma < c_5$ in Theorem 1 rules out any $c_5 > 5^{-5}$. In the usual density normalization, it also implies that $5!/5^5$ is a non-jump. The proof uses explicit finite hypergraphs whose Lagrangians approach 5^{-5} from above. We use the unnormalized Lagrangian of [4, Section 2],

$$P_H(x) = \sum_{e \in E(H)} \prod_{v \in e} x_v, \qquad \lambda(H) = \max_{\substack{x_v \geq 0 \\ \sum_v x_v = 1}} P_H(x), \tag{1.1}$$

without the factor $r!$; see also [11, 12].

Shaw's criterion [8, Theorem 3.1] uses a finite r-pattern P, whose edges may repeat vertices, and a vertex v with positive weight in a maximizing weighting x, such that $\lambda(\mathrm{FR}_v(P)) = \lambda(P) < 1$. For a simple P, the construction $\mathrm{FR}_v(P)$ replaces v by r clones and adds the edge consisting of those clones. Splitting x_v equally among them preserves the weight of the old edges and gives

$$\lambda(\mathrm{FR}_v(P)) \geq \lambda(P) + (x_v/r)^r > \lambda(P).$$

Thus the equality condition fails for each of the simple hypergraphs G_n used here. Instead, we bound their full Lagrangians as n tends to infinity.

Proposition 2. *For every integer $n \geq 2$, the 5-uniform hypergraph G_n defined in Section 2 has $2n + 7$ vertices and satisfies*

$$5^{-5}\left(1 + \frac{1}{4(3n)^3}\right) \leq \lambda(G_n) \leq 5^{-5} + \frac{1}{4!n}. \tag{1.2}$$

The mechanism is visible at the level of the cubic forms. After normalizing the bottom mass, the balanced weighting on the closed cycle has

$$b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{3},$$

so both cubic forms attain $1/27$ even with $D > 0$, while the additional edge $\{U, V\} \cup \mathcal{D}$ contributes a gain of order D^3. Deleting one vertex A_k opens the cycle; the terminal mismatch forces a loss of order D^3 with a coefficient large enough to absorb that gain. Consequently every open path T_L has Lagrangian exactly 5^{-5}, whereas the closed cycle has Lagrangian slightly above 5^{-5}. Since a long cycle always has a vertex of small weight, the passage from a cycle to a path also gives an upper bound approaching the same threshold.

2 The cyclic construction

Fix $n \geq 2$. Take pairwise distinct vertices

$$A_i, B_i \quad (i \in \mathbb{Z}/n\mathbb{Z}), \qquad C, U, V, \qquad W, \qquad D_1, D_2, D_3.$$

Put $\mathcal{D} = \{D_1, D_2, D_3\}$. On the vertex set $\{A_i, B_i : i \in \mathbb{Z}/n\mathbb{Z}\} \cup \{C\} \cup \mathcal{D}$, define two three-uniform hypergraphs by

$$E(H_1) = \bigcup_i \{\{A_i, X, Y\} : X \in \{B_i, C\},\ Y \in \{B_j : j \neq i\} \cup \mathcal{D}\},$$
$$E(H_2) = \bigcup_i \{\{B_i, X, Y\} : X \in \{A_{i+1}, C\},\ Y \in \{A_j : j \neq i+1\} \cup \mathcal{D}\}.$$

All subscripts are read modulo n. Define

$$\begin{aligned} E(G_n) = \{&\{W, U\} \cup e : e \in E(H_1)\} \\ &\cup \{\{W, V\} \cup e : e \in E(H_2)\} \cup \{\{U, V\} \cup \mathcal{D}\}. \end{aligned} \tag{2.1}$$

Each triple in H_1 has a unique A-vertex, which determines i; the disjoint sets of choices for X and Y then determine both vertices uniquely. The same argument applies to H_2, using its unique B-vertex. Thus these parameterizations count each triple once. Every edge of G_n has five distinct vertices, and the three edge families are distinguished by their intersections with $\{U, V\}$. Hence G_n is a simple 5-uniform hypergraph, with no repeated terms in the polynomial below.

Use lower-case letters for the weights of the corresponding vertices, and write

$$A = \sum_i a_i, \qquad B = \sum_i b_i, \qquad D = \sum_{j=1}^{3} d_j.$$

The two cubic forms and the polynomial of G_n are

$$P_1 = \sum_i a_i(b_i + c)(B - b_i + D), \tag{2.2}$$

$$P_2 = \sum_i b_i(a_{i+1} + c)(A - a_{i+1} + D), \tag{2.3}$$

$$P_{G_n} = w(uP_1 + vP_2) + uv \prod_{j=1}^{3} d_j. \tag{2.4}$$

Lemma 3. *The lower bound in* (1.2) *holds.*

Proof. Assign the weights

$$\begin{gathered} w = \frac{1}{5}, \qquad u = v = \frac{1}{10}, \qquad a_i = b_i = \frac{1}{5n}, \\ d_j = \frac{1}{15n}, \qquad c = \frac{n-1}{5n}. \end{gathered} \tag{2.5}$$

Their sum is one. Moreover,

$$A = B = c + D = \frac{1}{5}, \qquad b_i + c = B - b_i + D = a_{i+1} + c = A - a_{i+1} + D = \frac{1}{5}.$$

Thus $P_1 = P_2 = 5^{-3}$. The first term in (2.4) is 5^{-5}, and the second is

$$\frac{1}{100(15n)^3} = \frac{5^{-5}}{4(3n)^3}.$$

□

3 A bound for open paths

The path estimate uses the following inequality for finite sequences.

Lemma 4. *Let $N \geq 1$, and let $x_0, \ldots, x_N$ be nonnegative real numbers such that $x_N = 0$ and $\sum_{i=0}^{N-1} x_i \geq 1/2$. Then*

$$\sum_{i=0}^{N-1} x_i(1 - x_{i+1})^2 \geq \frac{2}{9}. \tag{3.1}$$

Proof. Define

$$\psi(x) = \begin{cases} x - x^2/2, & 0 \leq x \leq 1, \\ 1/2, & x > 1. \end{cases}$$

For all $x, y \geq 0$,

$$\psi(x) - \psi(y) \leq x(1 - y)^2. \tag{3.2}$$

For $x, y \leq 1$, this follows from the identity

$$x(1-y)^2 - \psi(x) + \psi(y) = \frac{(x - 2y + y^2)^2 + y(1-y)^2(2-y)}{2} \geq 0.$$

If $y \geq 1$, the left side of (3.2) is nonpositive. If $x > 1 > y$, it equals $(1-y)^2/2$, so the inequality holds in this case as well.

If some $x_j \geq 1/3$, summing (3.2) from j to $N-1$ gives

$$\sum_{i=0}^{N-1} x_i(1 - x_{i+1})^2 \geq \psi(x_j) - \psi(0) \geq \frac{5}{18} > \frac{2}{9}.$$

Otherwise every $x_i < 1/3$, and the sum is at least $\frac{4}{9}\sum_{i=0}^{N-1} x_i \geq 2/9$. $\square$

Lemma 5. *Let $L \geq 1$, and let $a_1, \ldots, a_L, b_1, \ldots, b_L, c, D$ be nonnegative numbers. Put*

$$A = \sum_{i=1}^{L} a_i, \qquad B = \sum_{i=1}^{L} b_i, \qquad a_{L+1} = 0,$$

and suppose that $A + B + c + D = 1$. Define the path forms

$$P_1 = \sum_{i=1}^{L} a_i(b_i + c)(B - b_i + D), \qquad P_2 = \sum_{i=1}^{L} b_i(a_{i+1} + c)(A - a_{i+1} + D).$$

Then, for every $s \in [0, 1]$,

$$sP_1 + (1-s)P_2 \leq \frac{1}{27} - \frac{s(1-s)D^3}{10}. \tag{3.3}$$

Proof. Set

$$\delta_A = A - \frac{1}{3}, \qquad \delta_B = B - \frac{1}{3}, \qquad S = |\delta_A| + |\delta_B|,$$

and

$$\alpha = \frac{A + D - c}{2} = D + \delta_A + \frac{\delta_B}{2}, \qquad \beta = \frac{B + D - c}{2} = D + \frac{\delta_A}{2} + \delta_B.$$

Completing the square gives

$$P_1 = \frac{A(1-A)^2}{4} - \sum_i a_i(b_i-\beta)^2, \tag{3.4}$$

$$P_2 = \frac{B(1-B)^2}{4} - \sum_i b_i(a_{i+1}-\alpha)^2. \tag{3.5}$$

Write

$$E = s\sum_i a_i(b_i-\beta)^2 + (1-s)\sum_i b_i(a_{i+1}-\alpha)^2, \qquad \Delta = \frac{1}{27} - sP_1 - (1-s)P_2.$$

The exact identity

$$\Delta = \frac{s\delta_A^2(4/3-A)}{4} + \frac{(1-s)\delta_B^2(4/3-B)}{4} + E$$

and the bounds $A, B \le 1-D$, $D \le 1$ imply

$$\Delta \ge \frac{D}{3}(s\delta_A^2 + (1-s)\delta_B^2) + E.$$

Indeed, $(4/3-A)/4 \ge (1/3+D)/4 \ge D/3$, and similarly for B. By the weighted Cauchy–Schwarz inequality,

$$\Delta \ge \frac{s(1-s)DS^2}{3} + E. \tag{3.6}$$

If $S \ge 11D/20$, then

$$\Delta \ge \frac{121}{1200}s(1-s)D^3 \ge \frac{s(1-s)D^3}{10},$$

as required. This includes $D=0$, since $S \ge 0$. Suppose therefore that $S < 11D/20$, which implies $D > 0$. Since $D \le 1/3 + S$, we have

$$D \le \frac{20}{27}, \qquad A + B \ge \frac{2}{3} - S \ge \frac{7}{27} > \frac{1}{4}, \qquad 0 < D - S \le \alpha, \beta \le \frac{1}{2}.$$

Apply Lemma 4 to the sequence

$$\frac{a_1}{\alpha}, \frac{b_1}{\beta}, \ldots, \frac{a_L}{\alpha}, \frac{b_L}{\beta}, 0.$$

Its sum is $A/\alpha + B/\beta \ge 2(A+B) > 1/2$. Each successive pair x, y contributes to E a term $s\alpha\beta^2 x(1-y)^2$ or $(1-s)\beta\alpha^2 x(1-y)^2$. Consequently

$$E \ge \frac{2}{9}\min\{s\alpha\beta^2, (1-s)\beta\alpha^2\} \ge \frac{2}{9}s(1-s)(D-S)^3. \tag{3.7}$$

For all $S, m \ge 0$ we have

$$\frac{(S+m)S^2}{3} + \frac{2m^3}{9} \ge \frac{(S+m)^3}{10}. \tag{3.8}$$

Multiplying the difference by 90 gives the nonnegative expression

$$\begin{aligned} &21S^3 + 3S^2m - 27Sm^2 + 11m^3 \\ &= 21\left(S - \frac{3m}{5}\right)^2\left(S + \frac{6m}{5}\right) + 3m\left(S - \frac{18m}{25}\right)^2 + \frac{233m^3}{625}. \end{aligned}$$

Taking $m = D - S$ in (3.8) and using (3.6)–(3.7) proves (3.3). □

For $L \ge 1$, define the 5-uniform hypergraph T_L as in (2.1), with each index ranging over $1, \dots, L$ and with the last link in H_2 replaced by the triples

$$\{B_L, C, Y\}, \qquad Y \in \{A_1, \dots, A_L\} \cup \mathcal{D}.$$

The edge $\{U, V\} \cup \mathcal{D}$ is retained. Its polynomial is (2.4) with $a_{L+1} = 0$.

Lemma 6. *For every $L \ge 1$, we have $\lambda(T_L) = 5^{-5}$.*

Proof. The lower bound follows by assigning weight $1/5$ to the vertices of any one edge and zero elsewhere. For the upper bound, take nonnegative vertex weights of total mass one, and put

$$h = w, \qquad q = u + v, \qquad t = A + B + c + D.$$

Thus $h + q + t = 1$. If $q = 0$ or $t = 0$, then $P_{T_L} = 0$. Otherwise put

$$s = u/q, \qquad d = D/t, \qquad z = s(1-s)d^3 \in [0, 1/4].$$

Applying Lemma 5 to the bottom weights divided by t, and then using homogeneity, gives

$$uP_1 + vP_2 \le qt^3 \left(\frac{1}{27} - \frac{z}{10} \right).$$

The arithmetic–geometric mean inequality now yields

$$P_{T_L} \le hqt^3 \left(\frac{1}{27} - \frac{z}{10} \right) + q^2 s(1-s) \left(\frac{td}{3} \right)^3. \tag{3.9}$$

Also,

$$\frac{qt^3}{27} \le \left(\frac{q+t}{4} \right)^4, \qquad \frac{q^2t^3}{3^3} \le 4 \left(\frac{q+t}{5} \right)^5. \tag{3.10}$$

Define

$$F(h) = \frac{3125}{256} h(1-h)^4.$$

We have $F(h) \le 1$, with equality at $h = 1/5$. Since $1 - 27z/10 > 0$, we may substitute (3.10) into (3.9). We get

$$\begin{aligned} \frac{P_{T_L}}{5^{-5}} &\le F(h) \left(1 - \frac{27}{10} z \right) + 4s(1-s)d^3(1-h)^5 \\ &= F(h) + z \left[4(1-h)^5 - \frac{27}{10} F(h) \right]. \end{aligned} \tag{3.11}$$

Since $0 \le 4z \le 1$, the last expression in (3.11) is the convex combination

$$(1-4z)F(h) + 4z \left[\left(1 - \frac{27}{40} \right) F(h) + (1-h)^5 \right].$$

As $F(h) \le 1$, it suffices to prove

$$\left(1 - \frac{27}{40} \right) F(h) + (1-h)^5 \le 1 \qquad (0 \le h \le 1). \tag{3.12}$$

Since $(1 - 27/40)(3125/256) = 8125/2048 < 5$, the binomial expansion gives

$$\left(1 - \frac{27}{40} \right) F(h) + (1-h)^5 \le 5h(1-h)^4 + (1-h)^5 \le \big(h + (1-h) \big)^5 = 1.$$

This completes the upper bound. $\square$

4 The Lagrangian bound and the blow-ups

Proof of Proposition 2. The lower bound was proved in Lemma 3. For the upper bound, take any nonnegative weights on G_n of total mass one. Choose an index k with

$$a_k \le \frac{A}{n} \le \frac{1}{n}.$$

After deleting A_k, put $a'_1 = a'_{n+1} = 0$, $a'_j = a_{k+j-1}$ for $2 \le j \le n$, and $b'_j = b_{k+j-1}$ for $1 \le j \le n$, with indices modulo n. Writing $A' = \sum_j a'_j$, $B' = \sum_j b'_j$, the remaining polynomial is

$$\begin{aligned} P_{G_n-\{A_k\}} &= wu \sum_{j=1}^{n} a'_j(b'_j + c)(B' - b'_j + D) \\ &\quad + wv \sum_{j=1}^{n} b'_j(a'_{j+1} + c)(A' - a'_{j+1} + D) + uv \prod_{\ell=1}^{3} d_\ell \\ &= P_{T_n}\big|_{a'_1=0}. \end{aligned} \tag{4.1}$$

Thus Lemma 6 and homogeneity bound the total weight of the edges that remain by

$$5^{-5}(1-a_k)^5 \le 5^{-5}.$$

For any simple 5-uniform hypergraph and weights of total mass one,

$$\frac{\partial P_H}{\partial x_v} \le \sum_{\substack{S \subseteq V(H)\setminus\{v\} \\ |S|=4}} \prod_{w \in S} x_w \le \frac{(\sum_{w \ne v} x_w)^4}{4!} \le \frac{1}{4!}. \tag{4.2}$$

Indeed, each product of four distinct weights occurs 4! times in the power expansion, and all other terms are nonnegative. The deleted edges consequently have total weight at most

$$a_k \frac{\partial P_{G_n}}{\partial a_k} \le \frac{a_k}{4!} \le \frac{1}{4!n}.$$

This proves (1.2). □

Proof of Theorem 1. Fix $\gamma > 5^{-5}$, and choose an integer $n \ge 2$ such that

$$5^{-5} + \frac{1}{4!n} < \gamma.$$

Keep n fixed, and set

$$\varepsilon = \frac{1}{4(3n)^3} > 0.$$

For each positive integer M, form a blow-up $B_{n,M}$ of G_n: replace each vertex by an independent class, and each edge by all transversal edges across its five classes. Use the following class sizes, with one class for each listed vertex:

vertex	W	U, V	A_i, B_i	C	D_j
class size	$6nM$	$3nM$	$6M$	$6(n-1)M$	$2M$

There are $N = 30nM$ vertices. These class sizes are proportional to the weights in (2.5), and the computation in Lemma 3 gives the exact edge count

$$e(B_{n,M}) = \left[(6n)^5 + 72n^2\right]M^5 = (1+\varepsilon)(N/5)^5. \tag{4.3}$$

Let S be any nonempty set of m vertices of $B_{n,M}$, and let k_v be the number of its vertices in the class corresponding to $v \in V(G_n)$. Then

$$e(B_{n,M}[S]) = \sum_{e \in E(G_n)} \prod_{v \in e} k_v = m^5 P_{G_n}\big((k_v/m)_v\big) \le m^5 \lambda(G_n) < \gamma m^5.$$

Together with (4.3) and $M \to \infty$, this proves the theorem. $\square$

Corollary 7. *For every integer $r \ge 5$ and every $\gamma > r^{-r}$, there is an $\varepsilon > 0$ and there are arbitrarily large r-uniform hypergraphs H such that*

$$e(H) \ge (1+\varepsilon)\big(v(H)/r\big)^r, \qquad e(H[S]) < \gamma |S|^r \quad \textit{for every nonempty } S \subseteq V(H).$$

In particular, Problem 1075 fails for every $r \ge 5$.

Proof. We implement Peng's lifting construction [6, proof of Theorem 1, p. 763] by adjoining vertices before taking a blow-up. Let $q = r - 5$. For a 5-uniform hypergraph G, form an r-uniform hypergraph G^{+q} by adjoining the same set of q new vertices to every edge. If the total weight on $V(G)$ is t, then the arithmetic–geometric mean inequality gives

$$P_{G^{+q}} \le \lambda(G) t^5 \left(\frac{1-t}{q}\right)^q,$$

with the usual interpretation when $q = 0$. The right side is maximized at $t = 5/r$, and equality is attained by taking an optimal weighting of G, scaled by $5/r$, and weight $1/r$ on each new vertex. Hence

$$\lambda(G^{+q}) = \frac{5^5}{r^r}\lambda(G). \tag{4.4}$$

Applying (4.4) to G_n and using Proposition 2 gives

$$r^{-r}\left(1 + \frac{1}{4(3n)^3}\right) \le \lambda(G_n^{+q}) \le r^{-r} + \frac{5^5}{r^r 4! n}.$$

Choose n so that the upper bound is less than γ. For each positive integer M, multiply every class size in the proof of Theorem 1 by 5, and give each of the q new vertices a class of size $30nM$. These are integer class sizes for a blow-up H of G_n^{+q}, with

$$\begin{aligned} N &= 30nrM, \\ e(H) &= 5^5\left[(6n)^5 + 72n^2\right](30n)^{r-5}M^r = (1+\varepsilon)(N/r)^r, \end{aligned} \qquad \varepsilon = \frac{1}{4(3n)^3}.$$

The induced-subgraph argument in the proof of Theorem 1 gives $e(H[S]) \le \lambda(G_n^{+q})|S|^r < \gamma|S|^r$. Letting $M \to \infty$ proves the claim. $\square$

Concluding remarks

The direct construction with $r-4$ common W-vertices and $r-2$ vertices in $\mathcal{D}$ has gain of order D^{r-2}; at $r=5$ this matches the cubic open-path loss, and Lemma 6 shows that the constants suffice to absorb it. The analogous power count at $r=4$ is heuristic: a quadratic gain would have to be absorbed by a cubic loss as $D \downarrow 0$. At $r=3$ the template does not exist, since a link edge $\{U\} \cup e$ already has four vertices. The endpoint question for $r=3,4$ remains open.

Together with Erdős's theorem that every number below $r!/r^r$ is a jump [2], Corollary 7 identifies $r!/r^r$ as the smallest non-jump for every $r \geq 5$. Mubayi raised in correspondence the question of whether this is the only transition for $r \geq 5$: are all numbers in $[r!/r^r, 1)$ non-jumps, or do jumps occur again above the first non-jump?

For $r=3$, Liu and Mubayi conjecture that every number in $[0, 4/9)$ is a jump and every number in $[4/9, 1)$ is a non-jump [9, Conjecture 1.2]. If this conjecture holds, the smallest non-jump for 3-graphs would be twice the Erdős threshold, whereas for every $r \geq 5$ it equals that threshold. Whether this dichotomy between $r=3$ and $r \geq 5$ is real, and what happens at $r=4$, is a natural next question.

Code availability

The Lean 4 formalization of Theorem 1 and Corollary 7, together with build instructions, is available at `https://github.com/FireflySentinel/erdos-1075`.

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Declaration of generative AI and AI-assisted technologies in the writing process

GPT-6 Astra proposed the cyclic construction and the Lagrangian estimates, and generated the manuscript and the Lean formalization. GPT-5.6 Sol and Claude Opus 5 were used for editorial review. The mathematical arguments, the Lean formalization and its correspondence with the manuscript were checked by the author, who takes responsibility for the content.

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